

BATCH: 2019-2013

FOR: Internship

NAME OF THE COMPANY: VISA Inc, Bangalore

ABOUT:

Visa Inc. is an American multinational financial services corporation headquartered in Foster City, California, United States. It facilitates electronic funds transfers throughout the world, most commonly through Visa-branded credit cards, debit cards and prepaid cards. They are a global payments technology company working to enable consumers, businesses, banks, and governments to use digital currency.

NUMBER OF ROUNDS: 2

1. Online Coding Round
2. Interview (Technical + HR)

ONLINE ROUND:

This round consists of two coding questions, given a duration of 60 minutes to solve it.

General tip: Since there are no MCQs, try to solve both the questions completely and focus more on corner cases.

The questions I was asked:

1. An investor has some money and wants to invest it in stock market. The number of stocks were given, and he can buy at most one share in any company. The total he invests should not exceed the funds available. A friend helps him make the maximum profit. Determine the maximum profit that can be earned.
Given: savings, currentValue array, futureValue array
Solution: same approach as 0-1 Knapsack. Here, n will be length of currentValue array, weights are the values of currentValue array, profits will be (futureValue – currentValue) and capacity is savings.
Reference link: <https://practice.geeksforgeeks.org/problems/0-1-knapsack-problem0945/1>
2. A scenario of defect product were given, solved similar to the following problem:
<https://www.geeksforgeeks.org/maximum-size-sub-matrix-with-all-1s-in-a-binary-matrix/>

A total of 17 students were shortlisted.

INTERVIEW:

The interview was 75 minutes long. Started with “introduce yourself” following which he told me about him, his achievements and a brief about VISA. Further I was asked about my projects and told to reason out my soft skills with examples.

After bringing me to the environment, we further moved to the coding part of the interview. We use hacker rank pair coding platform. I was asked to code two questions namely, reversing a linked list and implementation of a stack using two queues. Both the questions were told to be coded from scratch by defining structures and classes on our own and not the normal function definition we practice.

After implementing the code and running it for the told test cases, I was told to include constructors, destructors, and inheritance in my second question i.e., implement a stack using queues, to produce a working code and again the code was tested against given test cases.

With this the technical part was over and he further moved on to ask me DBMS questions. The questions I was asked include:

1. Writing simple SQL queries
2. Out of SQL, NoSQL, and Oracle databases, which is best and why
3. For RDBMS, which database would fit best?
4. Explain referential integrity
5. Explain the concept of indexing and its advantages
6. Difference between triggers and stored procedures

After this he asked me about critical section problems and the difference between threads and processes. With this the subject related questions were over and moved to product-based question.

The product-based question alone went to a discussion of 40 mins where I was asked to build an application like redBus. I was first asked to draw a layout for the app considering it to be a huge app, following which he told me to explain my ideas on implementation and building the app. The interviewer told me to assume him to be the customer and hence correctly split the application into three modules and give him one module at a time, further discussing about his requirements and making necessary changes before giving him the next module.

As the answer to the above I drew an ER diagram on the whiteboard and spoke about serverless architecture, AWS services such as lambda, S3 buckets and DynamoDB. I further included mailer functions, tokenization, and payments gateways. After presenting to him a consolidated layout I was asked to reduce the cost for which I removed transaction older than 5 days, removed unnecessary payment gateways and reduced number of tables used.

Lastly I was asked why I should be hired and with this the interview conclude.

At the end you will be asked if you have any questions, please do not say no, instead ask a question related to the company's achievement or new products.

TOPICS TO BE FOCUSED:

- Data Structure
- Database management system
- Operating system
- Whatever you've put in your resume
- OOPS

TAKE AWAY:

Having a good knowledge in CS basics and coding along with creative approaches for product-based questions will help in getting an internship in VISA.

Wishing you all the very best!

Shruthi Gopalakrishnan